

Bogeng Song 宋伯耕

Updated: June 2026

PhD Student, Psychology · Georgia Institute of Technology · Advisor: Dobromir Rahnev

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EDUCATION

PhD in Psychology — Georgia Institute of Technology Present
Advisor: Dobromir (Doby) Rahnev
Computational cognitive science & metacognition

MA in Psychology — New York University
Worked with Grace Lindsay, Marisa Carrasco

TRAINING & SUMMER SCHOOLS

NeuroAI — Neuromatch Academy Summer 2025
Deep learning as a framework for understanding brains and behavior

Computational Neuroscience — Neuromatch Academy Summer 2022
Models of neural dynamics, decision-making, and learning

PUBLICATIONS & PRESENTATIONS

† indicates equal contribution; underlined links are clickable.

Journal Articles

J2. R. Ezzo, B. Song, B. Rokers, & M. Carrasco. Eyes on hold: motion task difficulty jointly delays microsaccade and pupil responses. *Scientific Reports*. [Article](#) · [DOI](#) 2025

J1. X. Lu†, B. Song†, S. Zhang, S. Zhang, M. Huang, Y. Wang, & Y. Jiang. Implied gravity promotes coherent motion perception. *npj Microgravity*. [Article](#) · [DOI](#) 2025

Preprints

P1. N. C. Benson, B. Song, S. Chen, T. Miyata, H. Takemura, & J. Winawer. Machine learning matches human performance at segmenting the human visual cortex. *bioRxiv*. [Preprint](#) · [DOI](#) 2025

Conference Presentations

C5. B. Song, & D. Rahnev A Computational Signature of Self-Monitoring in Perceptual Confidence Cognitive Computational Neuroscience (CCN, 2026). 2026

C4. Q. Yang, H. W. Han, B. Song, J. Golomb, D. Rahnev, Y. Mohsenzadeh, & H.-H. Li. Hierarchical representational transformations of working memory in brains and machines. *Cognitive Computational Neuroscience (CCN, 2026)*. 2026

C3. B. Song, & D. Rahnev. The influence of response bias on confidence and accuracy in multi-alternative tasks for humans and artificial neural networks. *Vision Sciences Society (VSS,2025)*; *Journal of Vision*. 2025

C2. N. C. Benson, B. Song, T. Miyata, H. Takemura, & J. Winawer. Automated delineation of visual area boundaries and eccentricities by a CNN using functional, anatomical, and diffusion-weighted MRI data. *MODVIS (VSS satellite workshop)*. 2023

C1. R. Ezzo, B. Song, B. Rokers, & M. Carrasco. Microsaccade rates reflect trial difficulty for perifoveal motion discrimination. Vision Sciences Society (VSS); Journal of Vision.

2023

[Abstract](#)

COURSEWORK & COURSE PROJECTS

Differential contributions of episodic and semantic memory to story-telling — Computational Cognitive Modeling, NYU

ML classification of recalled / imagined / retold stories; feature importance linked to episodic vs. semantic memory theory.

Bayesian semi-supervised learning with function-space variational inference — Bayesian Machine Learning, NYU

Combined function-space variational inference with maximum-uncertainty regularization; evaluated on SVHN and CIFAR-10.

Other relevant coursework

Machine Learning (NYU, Center for Data Science) · Artificial Intelligence (Georgia Tech, School of Computer Science)

SKILLS

- Programming & web: Python, MATLAB, R, JavaScript, HTML, CSS
- Scientific computing & data analysis: NumPy, SciPy, Pandas, Matplotlib, statistical analysis, data visualization
- Machine learning & computational modeling: PyTorch, scikit-learn, deep neural network modeling, Bayesian methods, cognitive modeling, model fitting and simulation
- Behavioral and experimental methods: Psychophysics, eye-tracking, experimental design, behavioral data analysis, perceptual decision-making tasks
- Neuroimaging & NeuroAI: fMRI analysis, neural data analysis, visual-system modeling, representational analysis, deep networks as models of brain and behavior
- Tools & workflow: PsychoPy, Git/GitHub, LaTeX, Jupyter, reproducible research workflows
- Emerging AI interests: Large language models, vision-language models, reinforcement learning, human-AI comparison, learning and generalization